

# Backbone Chemistry Governs the Reliability of Machine-Learned Glass Transition Temperatures: A Family-Resolved Benchmark with Novelty-Conditioned Conformal Uncertainty

## Abstract

Machine-learned models of the glass transition temperature ( $T_g$ ) of homopolymers routinely report coefficients of determination above 0.85 and are increasingly proposed as screening tools for polymer discovery. Those figures come from random partitions, which fill the test set with repeat units whose backbone chemistry is already well represented in training; screening campaigns operate in the opposite regime. Using 7,174 canonical repeat units spanning twenty backbone families, we quantify what that difference costs, attribute it, and ask whether the uncertainty attached to a prediction survives it. Error rises from  $28.1 \pm 1.0$  K on random partitions to 33.8 K under scaffold holdout, 44.5 K under fingerprint-cluster holdout and 46.8 K under whole-family holdout, while  $R^2$  falls from 0.87 to 0.32. For scale, the two best models differ by 0.38 K, which repeated splits cannot resolve: **the evaluation protocol is worth about forty times more than the choice of model**. A matched-pair design, in which three models predict an identical held-out set from training pools differing only in what was removed, separates the two effects that a conventional family holdout confounds. Deleting an equal quantity of unrelated structures costs +0.32 K ( $p = 0.47$ ); deleting the family itself costs +14.61 K (95% CI [+10.53, +19.41],  $p = 2 \times 10^{-6}$ ) and is worse in all twenty families. The penalty is therefore caused by absent chemistry, not absent data, so it cannot be closed by collecting more of what is already covered. Transferability tracks structural isolation ( $r = -0.63$  to  $-0.68$ ,  $p < 0.005$ ) rather than family size ( $p > 0.2$ ), and spans  $0.80\times$  for polycarbonates to  $3.02\times$  for polyphosphazenes. Split conformal prediction fails in the same place: it holds 0.904 marginal coverage on random splits yet covers only 0.650 of the least familiar structural band, and loses even marginal validity under shift (0.834, 0.730, 0.711). Conditioning calibration on polymer family cannot repair this — a withheld family has no calibration members, so the predictor falls back for 100% of test structures and reproduces split conformal exactly — whereas conditioning on nearest-neighbour structural similarity holds 0.883–0.907 across all four regimes while remaining *narrower* than split conformal on random splits. We recommend reporting family-resolved errors and novelty-conditioned intervals, accompanied by the nearest-training similarity itself, alongside any  $T_g$  model intended for screening, and provide an open, fully reproducible implementation.

## 1. Introduction

The glass transition temperature is the single most consulted thermal descriptor of an amorphous polymer. It bounds the service window of a thermoplastic, sets processing conditions, and governs the mechanical response of a matrix in a composite. Measuring it by differential scanning calorimetry or dynamic mechanical analysis is routine but slow, and it requires a synthesised, purified sample, so predicting  $T_g$  from the structure of the repeat unit has been a target of polymer theory since group-contribution methods were first formalised [Bicerano].

Data-driven surrogates have largely displaced those schemes. Curated repositories such as PoLyInfo [Otsuka] made supervised learning on experimental  $T_g$  practical, and a decade of work has explored the resulting design space: physically interpretable descriptor sets with tree ensembles and kernel methods [Tao2021], graph neural networks over the repeat-unit graph [Qiu2023], and chemical language models pretrained on millions of hypothetical polymers [Kuenneth2023, Xu2023]. Recent community infrastructure has consolidated the datasets and the evaluation code [Jablonka2025, Xu2025]. The accuracies reported are genuinely useful: mean absolute errors in the region of 25–30 K against experimental values whose own inter-source spread can exceed 100 K.

Those numbers describe interpolation. They are obtained on random partitions, in which a test repeat unit typically has close structural analogues in training, and they therefore estimate how well a model fills in gaps within chemistry it has already seen. The motivating application is the reverse. A screening campaign enumerates candidates precisely because they are not yet in the literature, and the candidates that matter most are the ones furthest from the training distribution. The mismatch between the validation regime and the deployment regime is well known in cheminformatics, where scaffold and time-based splits were introduced for exactly this reason [Sheridan2013, Wu2018], and it has begun to be examined for polymers [Babbar2023, Akdogan2026].

Two questions remain open, and this work addresses both.

**Is the family-holdout penalty caused by the missing chemistry?** Withholding an entire polymer family and observing an error increase is the standard diagnostic, but it changes two things at once. The test population changes — families differ in how intrinsically hard they are and in how wide their  $T_g$  distribution is — and the training set shrinks. A comparison against a random-split baseline therefore measures the sum of a population effect, a data-volume effect and the effect of interest. Published treatments of this dataset state the confound explicitly and leave it unresolved. Resolving it requires holding the test set fixed and matching the training pools on size.

**Does the uncertainty survive the shift?** A point prediction without a trustworthy interval is not actionable for screening, and split conformal prediction [Papadopoulos2002, Vovk2005, Lei2018] is an attractive answer because it wraps any regressor in intervals with a finite-sample coverage guarantee under exchangeability. Two difficulties arise here. Exchangeability is precisely what a family holdout violates. And the guarantee is *marginal*: a predictor can achieve 90% coverage overall while systematically under-covering a subpopulation, and in this problem that subpopulation is the unfamiliar chemistry the model was deployed to explore. A recent analysis on a 410-sample simulation-derived set reports this pattern and characterises conformal intervals as conservative triage indicators rather than fine-grained screening tools [Akdogan2026]; whether the failure is repairable, and whether it persists at the scale and heterogeneity of the experimental record, has not been tested.

## 1.1. Contributions

- **An independently re-derived, chemically resolved dataset.** We re-curate

7,208 experimental repeat-unit records to 7,174 canonical structures, reproducing the published audit of this collection step for step, and assign each structure to one of twenty backbone families using a *backbone-aware* substructure classifier. Classifying on the chain path rather than on whole-molecule matches keeps poly(alkyl acrylate)s out of the polyester class, which matters because every family-holdout conclusion inherits the labels.

- **A matched-pair design that isolates the family effect.** Three training

pools — one retaining the family, one with it removed, and one size-matched control from which an equal number of unrelated structures was removed — predict an identical held-out test set, separating the cost of missing chemistry from the cost of missing data.

- **A conditional-coverage evaluation of conformal uncertainty,** and a

novelty-conditioned Mondrian taxonomy that restores subgroup validity where a family-conditioned taxonomy structurally cannot.

- **A validated applicability-domain criterion** expressed in units a

practitioner can act on: the nearest-neighbour Tanimoto similarity below which the stated interval should not be believed.

- **A reproducible open implementation,** with leak-free in-fold preprocessing

and a test suite pinning both the chemistry of the family assignment and the finite-sample behaviour of the conformal predictors.

## 2. Materials and Methods

### 2.1. Dataset and curation

We use the experimental homopolymer  $T_g$  collection redistributed with the POINT<sup>2</sup> polymer informatics benchmark [Xu2025], which derives from the PoLyInfo record [Otsuka]. Each entry pairs a repeat unit written as a PSMILES string — SMILES with two `\*` atoms marking the polymerisation attachment points — with a reported  $T_g$ .

The raw export is record-level: one repeat unit may appear several times carrying values from different primary sources. Modelling on records would allow identical inputs to fall on both sides of a partition and would weight frequently reported polymers more heavily, so all analysis is performed on a structure-level table. Repeat units were canonicalised with RDKit and grouped; where a canonical structure carried several reported values, the median was taken, and the number of contributing records and the full reported range were retained as provenance.

### 2.2. Backbone-aware family assignment

Polymer family labels were derived rather than inherited, so that the taxonomy is reproducible and its chemistry is inspectable. For each repeat unit the *backbone* is taken as the union of shortest paths between attachment points, extended to include any ring the path enters. Characteristic linkages are then matched against SMARTS patterns and accepted only when their core atoms lie on that backbone.

Backbone restriction is what separates a poly(alkyl acrylate) from a polyester: both contain an ester group, but in the acrylate it decorates an all-carbon chain. Patterns are tested in order of specificity, because the groups are nested — every cyclic imide contains two amide-like motifs, and a urethane contains both ester-like and amide-like fragments. Repeat units whose backbone carries no heteroatom linkage are classified by the groups pendant to the chain (aromatic-dominated, acrylic, styrenic, halogenated, other heteroatom-substituted vinyl, unsaturated, or saturated hydrocarbon). The twenty-two chemistry cases in Table S1 are pinned by unit tests.

### 2.3. Representations

Two representations were computed. The full RDKit descriptor block (217 columns) provides physically interpretable features; the topological `Ipc` index was replaced by its base-10 logarithm because it overflows for larger repeat units. Binary Morgan fingerprints (radius 2, 2,048 bits) provide the similarity space in which structural novelty is measured. Attachment points were retained in both, since they identify where the chain continues and the environments around them encode exactly the linkage motifs that set  $T_g$ .

Descriptor column filtering — removal of non-finite, constant and exactly duplicated columns — is implemented as the first step of a scikit-learn pipeline and is therefore **fitted on the training fold only**. Performing this filtering once over the full table, as is common, lets test structures influence the feature definition; the closest prior analysis of this dataset notes this as an unquantified limitation of its own results.

### 2.4. Partitioning regimes

Four regimes were used, in increasing order of enforced novelty: **random** partitions (repeated, 80:20); **scaffold** partitions, in which generic Bemis–Murcko scaffolds [Bemis1996] are assigned whole; **cluster** partitions, using average-linkage agglomerative clusters in Jaccard space over Morgan bits; and **family** holdouts, in which every member of one backbone family is withheld. All non-family regimes were repeated ten times with different seeds, because single-split rankings among tree ensembles on this dataset fall inside run-to-run noise.

## 2.5. Matched-pair design

For each eligible family  $F$ , a test set  $T \subset F$  is drawn and three training pools are constructed: **informed** (everything but  $T$ , so the remainder of  $F$  is available), **naive** (everything but  $F$  entirely), and **control** (**informed** minus a randomly chosen block of structures from other families, equal in number to  $F \setminus T$ ). All three predict the same  $T$ .

The contrast **naive** – **control** is the quantity of interest. Both pools are identical in size and both exclude  $T$ ; they differ only in *which* structures were removed. The contrast **control** – **informed** measures the pure data-volume effect over the same test set.

## 2.6. Models

Five regressors were evaluated on the descriptor representation: a training-median baseline, support vector regression, random forest, histogram gradient boosting and extremely randomised trees. Every model is wrapped in the same pipeline so that all data-dependent preprocessing is fitted in-fold.

## 2.7. Conformal prediction and coverage diagnostics

Each training pool is divided three ways: a **fit** partition trains the point regressor, a **difficulty** partition supplies out-of-sample residuals, and a **calibration** partition supplies the conformal quantile. Fitting the difficulty model on the point model's own training residuals would teach it that the model is everywhere accurate, so that partition must be held out. All conformal variants share one point predictor and one calibration set, so differences between them reflect only how residuals are converted into intervals.

Given calibration scores  $s_1 \dots s_n$  and a miscoverage level  $\alpha$ , the conformal quantile is the  $[(n+1)(1-\alpha)]$ -th order statistic; when that index exceeds  $n$  the honest interval is unbounded. We report  $\alpha = 0.1$  throughout. Three predictors are compared:

- **SCP** — one global quantile of absolute residuals.
- **Normalised SCP** — residuals scaled by a fitted difficulty estimate, so intervals widen where the model expects to struggle rather than everywhere.
- **Mondrian SCP** — a separate quantile per category [Vovk2012], under two taxonomies: *polymer family*, and *nearest-training-similarity band*.

The asymmetry between the two taxonomies is the point. A family-conditioned taxonomy is undefined for a family with no calibration members, which is exactly the family-holdout case; a novelty-conditioned taxonomy is defined for any repeat unit, seen family or not.

Alongside marginal coverage and mean interval width we report **conditional** coverage within each similarity band and each family, the **maximum coverage gap** (the largest shortfall below nominal across subgroups with at least ten members), and the coverage-width criterion. Ensemble spread is additionally assessed by sparsification error.

## 2.8. Reproducibility

All code, the curated structure-level table and every result file are available at [repository]. Seeds are fixed; a cold run reproduces every number in this paper.

# 3. Results

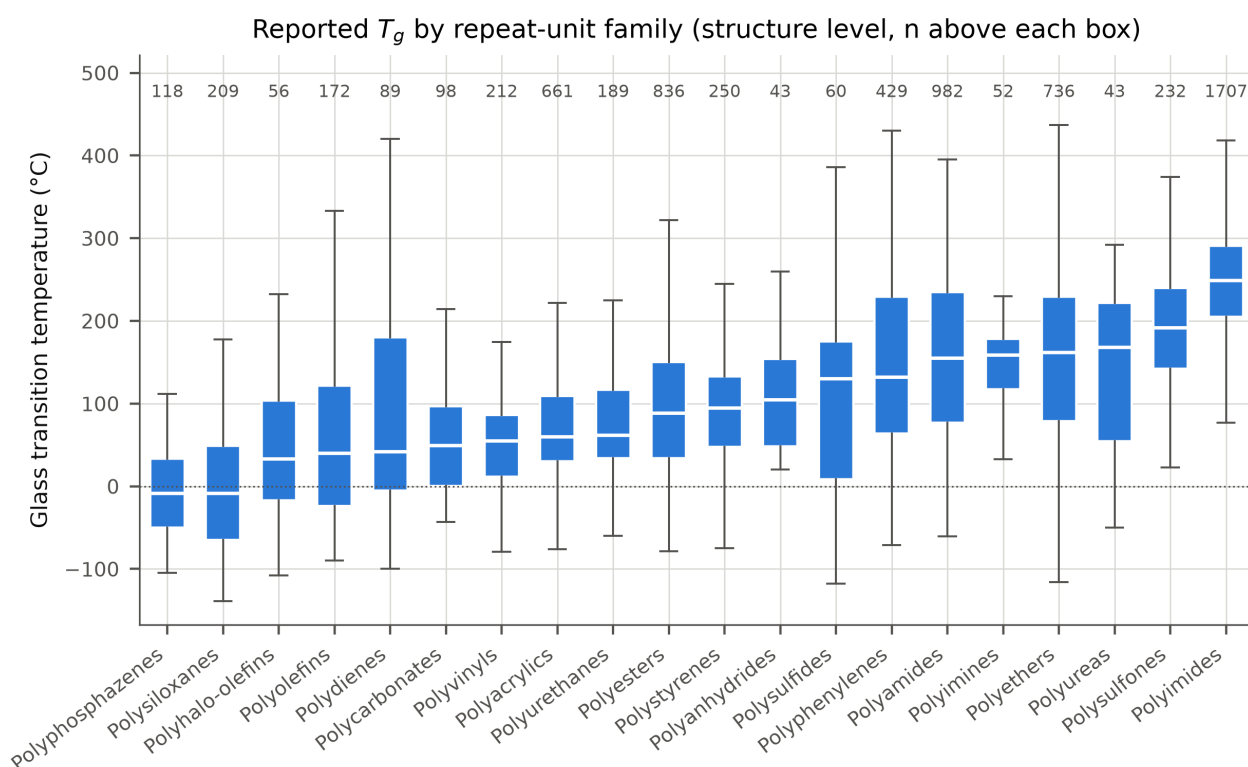
## 3.1. The public record survives re-derivation

Independent re-curation of the 7,208 raw records reproduces the published audit of this collection step for step: 7,174 unique raw SMILES, no RDKit parse failures, 7,174 canonical structures, 31 canonical groups containing more than one record, 34 records absorbed by aggregation, 28 of those groups carrying disagreeing  $T_g$  values, and a maximum within-structure

range of 115 K. Attachment-point counts likewise agree, at 7,170 structures with two, three with three and one with four. The dataset is therefore a stable object to build on, and the numbers below are not sensitive to curation choices.

That 115 K figure deserves emphasis before any model error is quoted. It is the spread between independent literature reports of the *same* repeat unit, and it places a floor under what structure-based prediction can achieve: a repeat-unit representation cannot resolve differences in molecular weight, tacticity, crosslinking, thermal history or measurement protocol, and those differences are folded into the target.

The derived taxonomy places all twenty-five reference polymers of Table S1 in the family a polymer chemist would assign. The resulting family  $T_g$  ordering (Figure 1) is likewise the expected one, which is a further check that the labels are chemically meaningful rather than merely self-consistent: polyphosphazenes (median  $-8^\circ\text{C}$ ) and polysiloxanes ( $-8^\circ\text{C}$ ) sit lowest, reflecting the low rotational barriers of P=N and Si-O backbones, and aromatic polyimides sit highest ( $249^\circ\text{C}$ ), reflecting rigid, strongly interacting heteroaromatic chains.



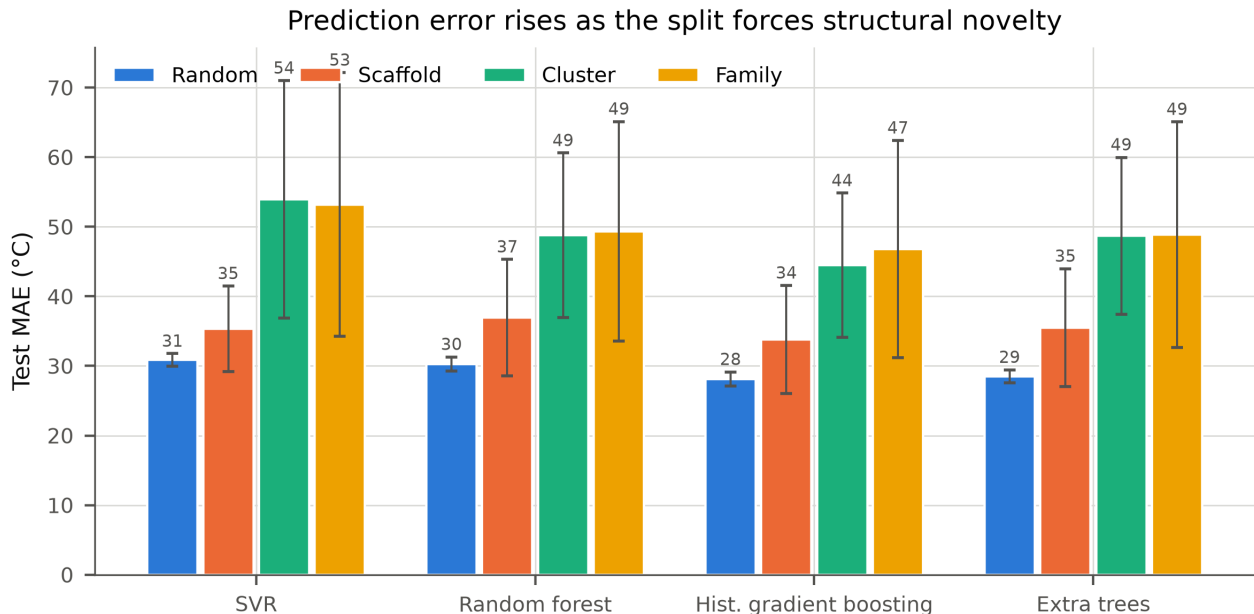
**Figure 1.** Reported glass transition temperature by repeat-unit family, ordered by median. Family sizes are printed above each box. The ordering is the one polymer chemistry predicts, which is evidence that the derived taxonomy is meaningful rather than merely self-consistent.

### 3.2. Accuracy is governed by the split, not by the model

On random partitions the best model reaches  $28.1 \pm 1.0$  K MAE ( $R^2 = 0.87$ ), consistent with published values for descriptor-based ensembles on this dataset. That ranking is fragile. Across ten repeated splits, histogram gradient boosting beats extremely randomised trees by 0.38 K, which a paired  $t$ -test does not resolve ( $p = 0.069$ ); random forests and support vector regression are worse by 2.17 K and 2.74 K ( $p < 0.0001$  each). Reporting a single split, as is common, would allow either of the leading models to be declared the winner.

Against that 0.38 K of model-choice sensitivity, the choice of partition moves the error by an order of magnitude more (Figure 2, Table 3). Holding the model fixed, MAE rises from 28.1 K on random splits to 33.8 K on scaffold splits, 44.5 K on fingerprint-cluster splits and 46.8 K under whole-family holdout;  $R^2$  falls from 0.87 to 0.79, 0.54 and 0.32. Mean nearest-training Tanimoto

similarity falls alongside it, from 0.76 on random splits to 0.67, 0.40 and 0.46. That ordering is not strictly monotone: family holdout leaves test structures slightly *less* isolated on average (0.46) than cluster holdout (0.40), because a withheld family may still have near analogues in a chemically adjacent one, yet family holdout produces the higher error. Structural distance alone therefore does not determine difficulty; *which* chemistry is missing also matters, a point Section 3.3 takes up directly. **The evaluation protocol is worth about forty times more than the model choice** (16.3 K against 0.38 K), which reframes what a leaderboard on this dataset is measuring.



**Figure 2.** Test MAE by model and split regime; error bars are the standard deviation across repeated splits. The spread between regimes dwarfs the spread between models.

The dispersion is as informative as the mean. Random-split MAE varies by  $\pm 1.0$  K across repeats; scaffold and cluster splits vary by  $\pm 7.8$  K and  $\pm 10.4$  K, and family holdouts by  $\pm 15.6$  K. Performance under novelty is not a single number, and a point estimate from one chemistry-aware split should not be trusted.

### 3.3. Transfer failure is a property of backbone chemistry

Family-holdout deterioration, defined as the ratio of family-holdout MAE to random-split MAE for the same model, spans a factor of nearly four across the twenty families (Table 4). At the difficult end sit polyphosphazenes ( $3.02\times$ , 84.8 K MAE), polydienes ( $2.58\times$ ) and polysiloxanes ( $2.48\times$ ). At the easy end, polycarbonates deteriorate by  $0.80\times$  — that is, they are predicted *better* when their own family is withheld than the average random-split structure is.

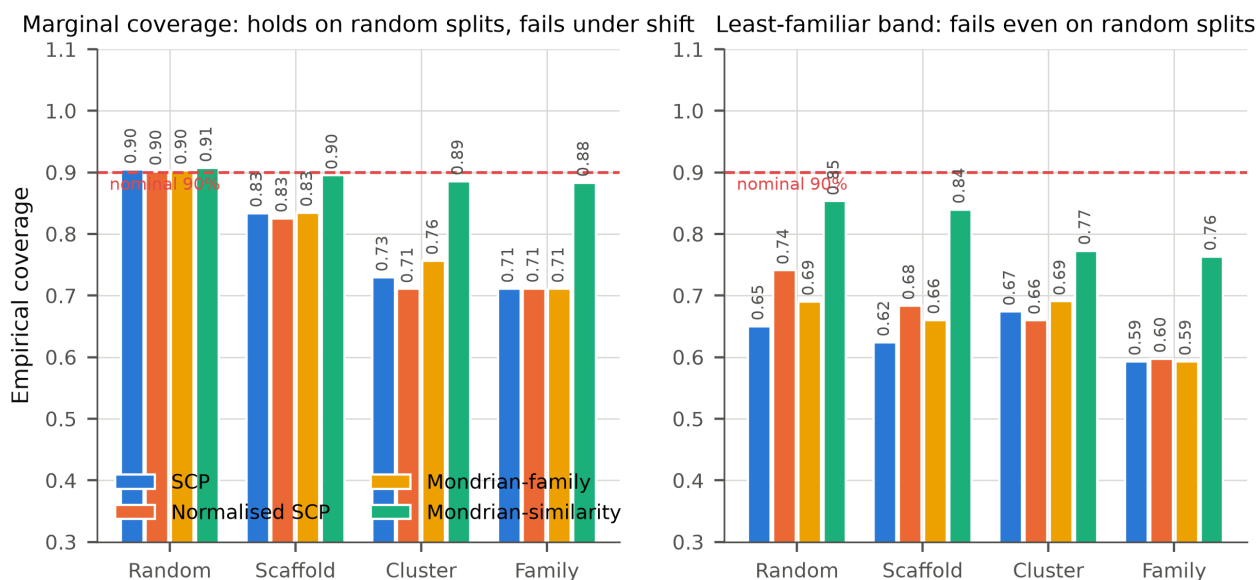
The pattern is chemically coherent. Polyphosphazene and polysiloxane backbones are inorganic, and nothing else in a dataset dominated by organic condensation polymers constrains the relationship between their descriptors and chain flexibility; the models fall back on organic intuition and overpredict badly. Polycarbonates, by contrast, are surrounded in structure space by polyesters and polyethers, which remain in training and carry nearly the same carbonyl and ether linkages, so their withdrawal removes little the model cannot recover.

Testing three candidate explanations against deterioration across the twenty families, only structural isolation survives. Mean nearest-training similarity correlates strongly and significantly for every model (Pearson  $r$  between  $-0.63$  and  $-0.68$ ,  $p < 0.005$ ). The shift between the family's median  $T_g$  and the rest of the dataset is weaker and marginal ( $r \approx 0.40$ – $0.46$ ,  $p \approx 0.04$ – $0.08$ ), and family size does not predict deterioration at all ( $p > 0.2$ ). Larger families are not easier to extrapolate to; isolated ones are harder.

This also settles a practical question. Because the effect tracks structural isolation rather than family size, collecting more data *within* well-represented families will not close the gap. Coverage of the structure space, not volume, is the binding constraint.

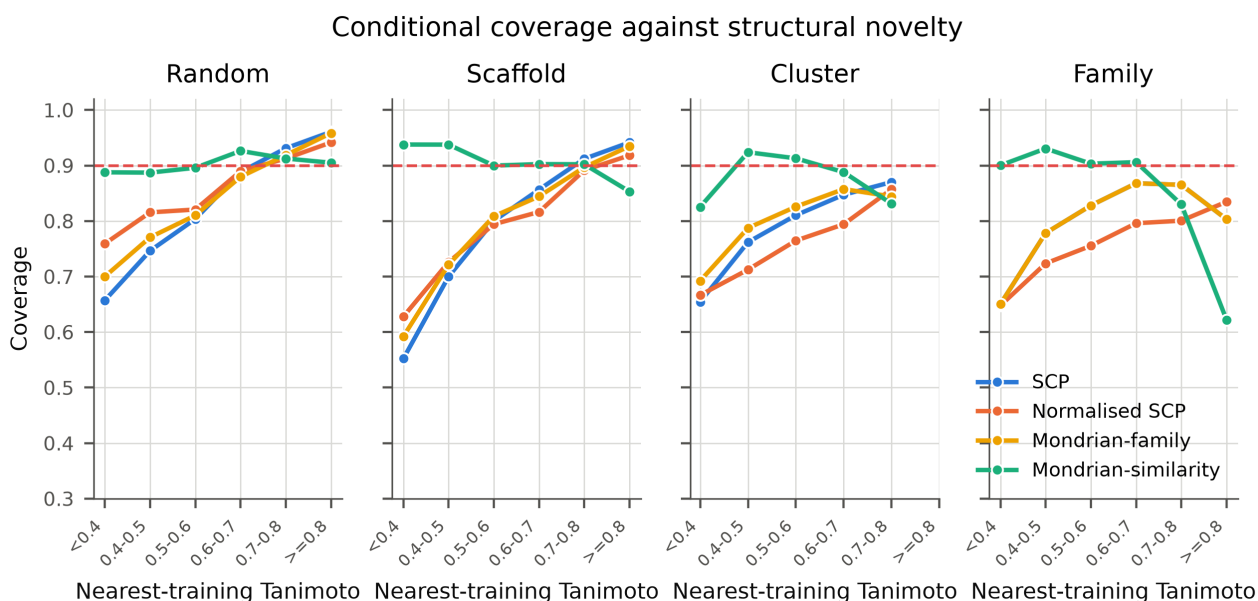
### 3.4. Conformal intervals lose validity exactly where they are needed

Split conformal behaves as advertised on random partitions: 0.904 empirical coverage against a nominal 0.90. Under shift the guarantee degrades monotonically with structural novelty — 0.834 on scaffold splits, 0.730 on cluster splits and 0.711 under family holdout (Figure 4, left). This is not a defect of the method but of its premise: conformal validity requires calibration and test data to be exchangeable, and a chemistry-aware holdout is constructed precisely to break that.



**Figure 4.** Coverage of nominally 90% conformal intervals. Left: marginal coverage. Right: coverage in the least-similar structural band of each split.

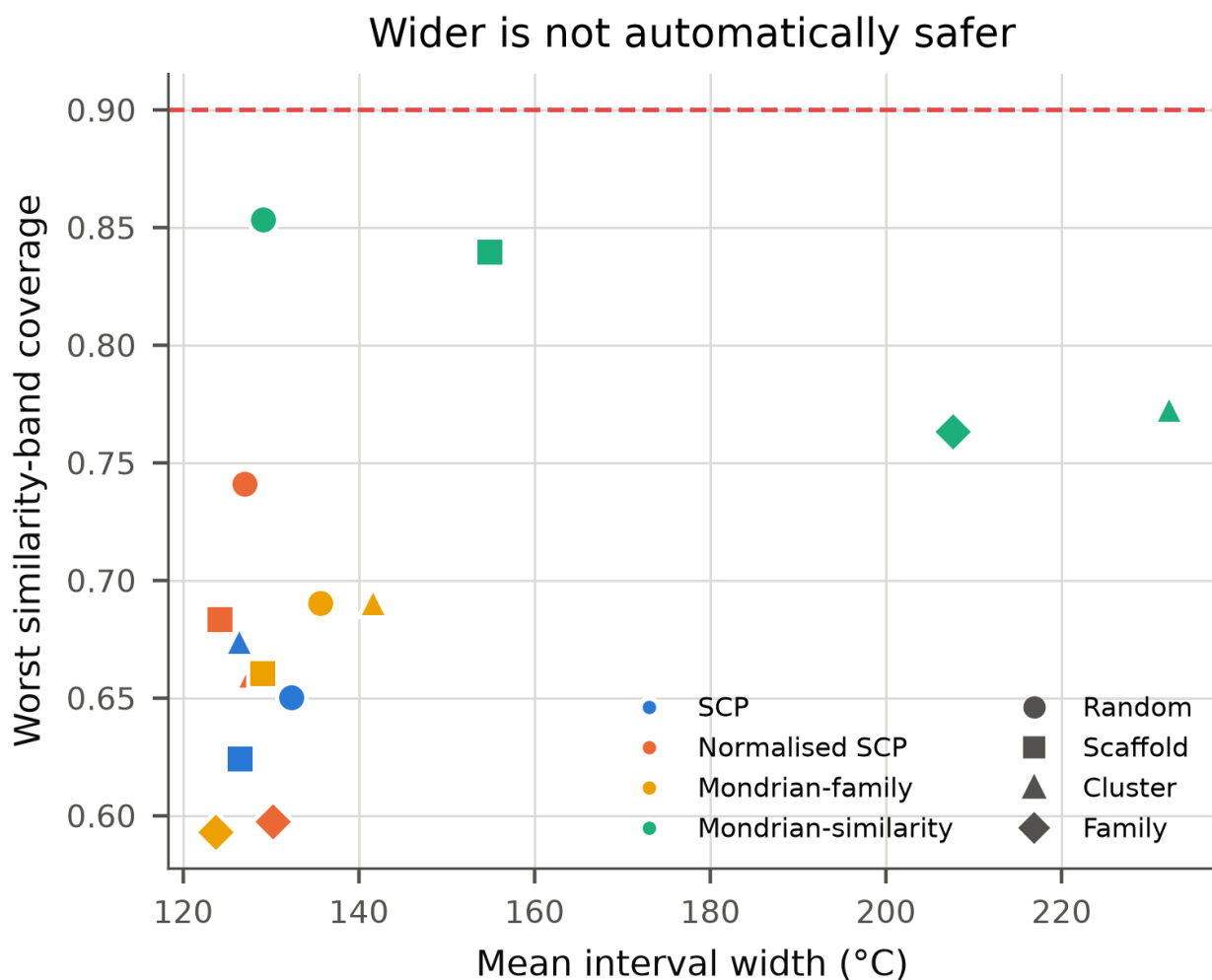
Marginal coverage also conceals a failure that is present even when it holds. On random splits, where overall coverage is a healthy 0.904, coverage in the least-similar structural band is 0.650 (Figure 4, right; Figure 5). A nominally 90% interval covers two-thirds of the repeat units that a screening campaign would actually be evaluating. **A model can pass every validity check in current practice and still be wrong about a third of the unfamiliar polymers it is deployed on.**



**Figure 5.** Conditional coverage against nearest-training Tanimoto similarity, one panel per split regime. Split conformal degrades monotonically with structural novelty; the novelty-conditioned quantile stays near nominal.

Neither obvious remedy works. The normalised predictor, which scales residuals by a fitted difficulty estimate, improves the worst band on random splits (0.741 vs 0.650) but fails with everything else under shift (0.826 scaffold, 0.711 cluster, 0.712 family) and is the *worst* of the four under cluster holdout. Conditioning the quantile on polymer family fails for a structural reason rather than an empirical one: under family holdout the withheld family has no calibration members by construction, so the predictor falls back to the global quantile for 100% of test structures and reproduces split conformal exactly (0.711, identical interval widths).

Conditioning on nearest-training similarity is the variant that survives, because that coordinate is defined for every repeat unit whether or not its family was seen. It holds 0.907, 0.895, 0.886 and 0.883 across the four regimes, never requiring a fallback under family holdout. Crucially it does not buy this with width: on random splits it is *narrower* than split conformal (129.1 K vs 132.3 K) while covering far more of the difficult band, and it widens only where the chemistry demands it — to 155 K under scaffold, 232 K under cluster and 208 K under family holdout. That adaptivity is the behaviour a screening workflow needs, and it is what the coverage-width criterion rewards (Figure 6): under family holdout, 342 against 35,556 for split conformal.



**Figure 6.** Worst similarity-band coverage against mean interval width. Marker colour is the conformal method, shape the split regime.

### 3.5. An applicability domain, and an honest limit on the repair

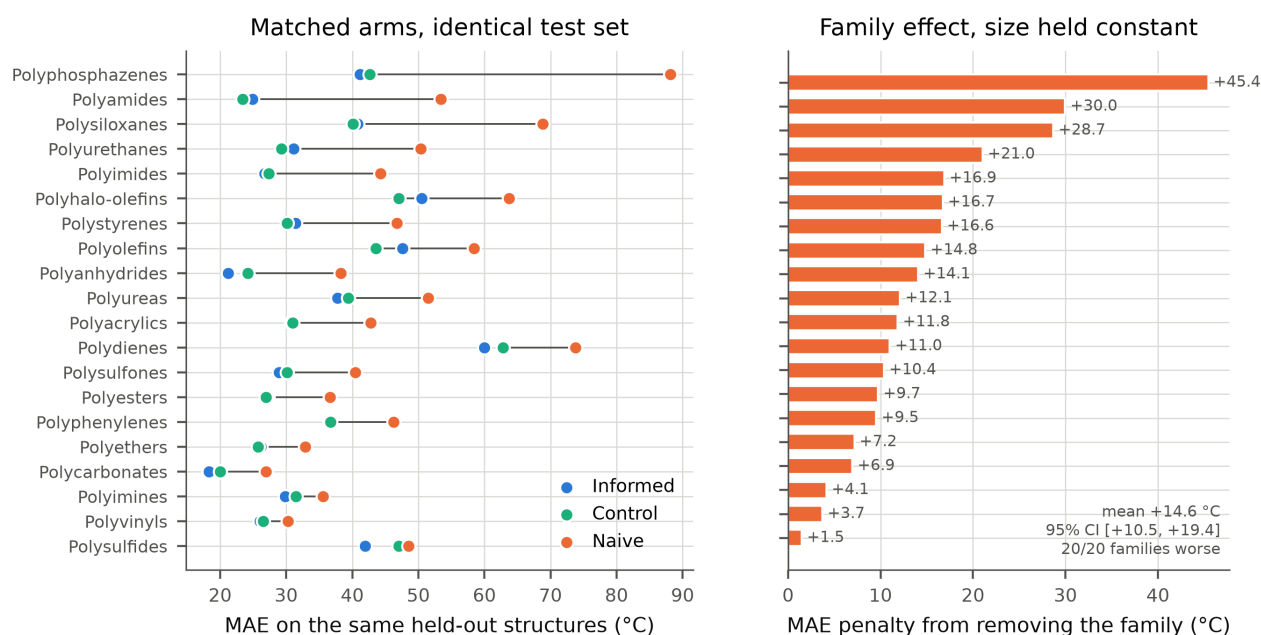
Resolving coverage by similarity band under family holdout (Table 5) turns the result into an operational rule. Split conformal covers 0.650 of structures whose nearest training analogue lies below Tanimoto 0.4, rising through 0.778, 0.828 and 0.868 as similarity increases. Its intervals are essentially constant in width (123–127 K) across all six bands — it applies one width to structures whose error differs by a factor of three, which is precisely the failure mode. Novelty-conditioned calibration instead holds 0.900, 0.931, 0.903 and 0.906 in the four least-familiar bands, at widths that scale from 250 K down to 158 K.

The repair has a cost that should be stated plainly. In the most-similar band (Tanimoto  $\geq 0.8$ ) the novelty-conditioned predictor *under-covers*, at 0.622 against split conformal's 0.803, because it narrows those intervals to 86 K. Under a family holdout a test structure that closely resembles training data is unusual, and the calibration points populating that band come from families where high similarity genuinely did signal an easy prediction. The trade is nonetheless strongly favourable in practice: that band holds 127 of 7,174 held-out predictions (1.8%), while the two least-similar bands hold 4,759 (66%). The recommendation is therefore to report novelty-conditioned intervals together with the nearest-training similarity itself, so that a reader can see which regime a given prediction sits in rather than trusting a single interval uniformly.

### 3.6. The penalty is caused by the missing chemistry, not the missing data

The matched design separates the two effects that a conventional family holdout confounds (Figure 3, Table 6). Across twenty families and three repeats, with the test set held identical and

the two comparison pools held identical in size:



**Figure 3.** Matched-pair design. Left: the three training arms predicting an identical set of held-out structures, with informed and control almost coincident. Right: the family effect with training-set size held constant.

- Removing an equal quantity of *unrelated* structures — the **control** arm —

changes MAE by +0.32 K (95% CI [−0.63, +1.23], Wilcoxon  $p = 0.47$ , worse in 12 of 20 families) for extremely randomised trees, and by +0.21 K ( $p = 0.93$ , 9 of 20) for histogram gradient boosting. Per family the effect ranges from −4.0 K to +5.1 K and scatters around zero.

- Removing *the family itself* — the **naive** arm — costs +14.61 K

(95% CI [+10.53, +19.41],  $p = 2 \times 10^{-6}$ ) and **is worse in all twenty of twenty families**, and +13.28 K ( $p = 1 \times 10^{-5}$ , 18 of 20) for the second model.

The family effect is roughly forty-six times the size of the data-volume effect and, unlike it, is consistent in sign across every family tested. Essentially the entire family-holdout penalty is attributable to the absent chemistry: the total effect of +14.93 K decomposes into +14.61 K of chemistry and +0.32 K of data volume. This resolves the confound that motivated the design, and it does so on matched test structures, so it is not an artefact of families differing in intrinsic difficulty or in  $T_g$  spread.

Family-level magnitudes rank as the chemistry predicts. Polyphosphazenes suffer most (+45.4 K), followed by polyamides (+30.0 K) and polysiloxanes (+28.7 K); at the other end polysulfides (+1.5 K), polyvinyls (+3.7 K) and polyimines (+4.1 K) are nearly unaffected, because the linkages that set their backbone mobility survive elsewhere in the training set. Note that polyamides rank second here despite being the second-largest family in the dataset (982 structures) — further evidence that abundance does not confer transferability when the chemistry is distinctive.

The practical reading is uncomfortable but clear. Because the penalty comes from absent chemistry rather than absent volume, a  $T_g$  model cannot be made trustworthy on a new backbone family by training it on more of what it already has. Either the family must be represented in training, or the prediction must carry an interval that widens honestly when it is not — which is the case for novelty-conditioned calibration made in Section 3.4.

## 4. Discussion

### 4.1. What a $T_g$ leaderboard is actually measuring

The most consequential number in this study is not an error but a ratio. Changing the model changes MAE by 0.38 K, an amount ten repeated splits cannot resolve; changing the partitioning protocol changes it by 16.3 K. Any comparison of architectures on this dataset that does not fix and report the split regime is dominated by a variable it does not control.

This reframes a decade of incremental gains. Reported improvements from kernel methods to tree ensembles to graph networks to pretrained chemical language models are real, but they are improvements in interpolation, measured where structural analogues are plentiful. None of the comparisons establishes that the newer model extrapolates better, because none was evaluated under a regime that requires extrapolation. That is not a criticism of the models; it is a criticism of the protocol they were scored under, and it is cheap to fix.

## 4.2. Why chemistry and not volume

The matched design's central result — +14.61 K for removing a family against +0.32 K for removing the same number of unrelated structures — has a straightforward physical reading. Descriptor-based models of  $T_g$  learn how particular backbone linkages translate into chain stiffness and interchain interaction. Those mappings are local to a linkage type. Remove every phosphazene, and no amount of additional polyester data teaches the model what a P=N backbone does to segmental mobility; it extrapolates organic intuition into inorganic chemistry and overpredicts by tens of kelvin.

The correlation analysis supports the same reading from a different direction. Deterioration tracks nearest-training similarity but not family size, and polyamides — the second-largest family in the dataset at 982 structures — rank second worst in the matched design. A family can be simultaneously abundant and untransferable if its characteristic linkage is not reproduced elsewhere.

This has a direct consequence for data-collection strategy. The instinct when a model underperforms is to gather more data; these results say that gathering more data *within* already-covered chemistry will not help at all with new backbones. Coverage of the structure space, not dataset size, is the binding constraint, and a modest number of well-chosen structures from an uncovered family is worth more than a large number from a covered one.

## 4.3. Why novelty-conditioning works and family-conditioning cannot

The contrast between the two Mondrian taxonomies is not an empirical accident but a structural fact, and it generalises beyond this dataset. A conditional conformal predictor can only offer category-specific validity for categories it has calibration data for. Polymer family fails this test in exactly the deployment scenario of interest: the family is new, so it has no calibration members, and the predictor silently degrades to the unconditional one. Our family-holdout experiments show this happening for 100% of test structures, with coverage and interval widths identical to split conformal to three decimal places.

Nearest-training similarity has no such blind spot. It is computable for any repeat unit against any training set, its value is available before the label is, and its bands are populated by calibration structures drawn from every family. The taxonomy therefore remains well defined precisely where the family-based one collapses. The same argument recommends similarity — or any other deployment-computable novelty coordinate — over class-based taxonomies for conditional conformal prediction in materials problems generally, wherever the classes of interest are the ones the model has not seen.

It is worth stressing that the improvement is not bought with width. On random splits the novelty-conditioned predictor is narrower than split conformal (129.1 K vs 132.3 K) while covering the difficult band far better. It widens only where the chemistry warrants, reaching 232 K under cluster holdout. An interval predictor that simply inflated everything would score well on coverage and be useless for ranking candidates; this one reallocates width from the easy structures to the hard ones.

#### 4.4. Practical recommendations

For anyone building or reporting a  $T_g$  model intended for screening:

- **Report the split regime as a primary result, not a methods detail**, and report at least one chemistry-aware regime alongside any random split. The gap between them is the honest estimate of screening performance.
- **Repeat every regime**. Chemistry-aware splits vary by  $\pm 8$ –16 K across repeats; a single number from one scaffold split is not interpretable.
- **Report errors resolved by polymer family**, not only in aggregate. An aggregate MAE of 47 K conceals a range from 22 K to 85 K, and a user working on polysiloxanes is served by the wrong number.
- **\*\*Ship intervals, and condition them on a deployment-computable novelty coordinate.\*\*** Report the nearest-training similarity next to each prediction so a reader can see which regime it belongs to.
- **Fit every data-dependent preprocessing step inside the fold**, including descriptor filtering. It is a one-line change to a pipeline and removes an unquantified optimism from the reported numbers.

#### 4.5. Limitations

The target itself is noisy in a way no structural model can address. Independent literature reports of the same repeat unit differ by up to 115 K in this dataset, because a repeat-unit graph cannot express molecular weight, tacticity, crosslink density, thermal history or measurement protocol. A 28 K random-split MAE should be read against that floor, and part of the residual is irreducible.

The family taxonomy is our own. It is rule-based, inspectable and validated against twenty-five reference polymers (Table S1), which we prefer to an opaque inherited label, but a different chemically defensible taxonomy would shift the family-level numbers. The matched-pair conclusion is robust to this in kind if not in magnitude, since it holds for all twenty families individually rather than only on average.

Our models are descriptor-based ensembles. We show in Section 3.7 that the generalisation gap survives a change of representation to Morgan fingerprints, but we have not tested a pretrained chemical language model or a graph neural network. Such a model, having seen millions of hypothetical polymers during pretraining, might have effective coverage of families absent from this dataset's *labelled* portion. Testing whether pretraining narrows the family effect is the natural next experiment, and the matched design transfers to it unchanged.

Finally, the novelty-conditioned predictor under-covers the most-similar band under family holdout (Section 3.5). The affected group is small (1.8% of predictions) and the direction of the trade favours screening, but the method is an improvement rather than a solution, and conditional coverage should continue to be reported rather than assumed.

### 5. Conclusions

Machine-learned glass transition temperatures are far more reliable inside familiar backbone chemistry than outside it, and standard practice measures only the former. On 7,174 experimental repeat units, moving from random to chemistry-aware partitions raises error from 28.1 K to as much as 46.8 K and lowers  $R^2$  from 0.87 to 0.32 — a shift roughly forty times larger than the difference between the best and second-best model.

A matched-pair design shows that this penalty is caused by the absent chemistry rather than the absent data: removing an equal quantity of unrelated structures costs +0.32 K and is statistically

indistinguishable from zero, while removing the family costs +14.61 K and is worse in all twenty families tested. Transferability tracks how structurally isolated a family is, not how large it is, so it cannot be bought with more data from chemistry that is already covered.

The uncertainty attached to these predictions fails in the same place. Split conformal intervals hold their nominal 90% marginally on random splits while covering only 65% of the least familiar structures, and lose marginal validity entirely under shift. Conditioning the calibration on polymer family cannot repair this, because a withheld family has no calibration data by construction. Conditioning on nearest-training structural similarity does, holding 0.88–0.91 across every regime tested without inflating intervals on easy predictions.

The practical implication is a short list: report a chemistry-aware split, repeat it, resolve errors by family, and attach novelty-conditioned intervals together with the similarity value itself. None of these is expensive, and together they turn a  $T_g$  model from something that scores well into something a screening campaign can act on.

### 3.7. The gap is not an artefact of the descriptor set

A natural objection is that the generalisation gap reflects a limitation of RDKit descriptors rather than of the data, and that a representation better suited to novel chemistry would close it. Repeating the random and family-holdout regimes with binary Morgan fingerprints (radius 2, 2,048 bits) in place of the descriptor block tests this directly, since the two representations share almost nothing: one is a set of physically motivated aggregate quantities, the other a sparse record of local substructural environments.

The gap survives the substitution, and widens slightly (Table 7). Fingerprints are worse in absolute terms in both regimes — 35.8 K against 28.5 K on random splits and 63.7 K against 48.9 K under family holdout — and their family-holdout penalty is marginally *larger*, both in absolute terms (+27.9 K against +20.4 K) and as a ratio (1.78× against 1.71×). Substructural environments offer no protection here, which is consistent with the mechanism proposed in Section 4.2: the bits encoding a P=N or Si-O backbone simply do not occur in the training data when that family is withheld, so a fingerprint model has even less to generalise from than a descriptor model whose aggregate quantities at least vary continuously.

This does not establish that *no* representation could narrow the gap. A model pretrained on millions of hypothetical polymers may have effective coverage of backbone chemistry that this dataset's labelled portion lacks, and testing that is the natural next step (Section 4.5). It does establish that the gap is a property of which chemistry is present in the training data, not of the particular feature set used here.

### 3.8. The calibration set is a second, independent lever

Sections 3.4 and 3.5 repair conditional coverage by changing how the conformal *quantile* is computed. A second possibility is to leave the quantile alone and change what the calibration set contains. Standard split conformal draws calibration points at random from the training pool, so every calibration residual describes an interpolation — the model had relatives of that repeat unit in training. Under family holdout the test residuals describe extrapolation, so the quantile is calibrated against the wrong population. Rebuilding the calibration set by holding families out *within* the training pool produces residuals of the kind the test set will actually demand.

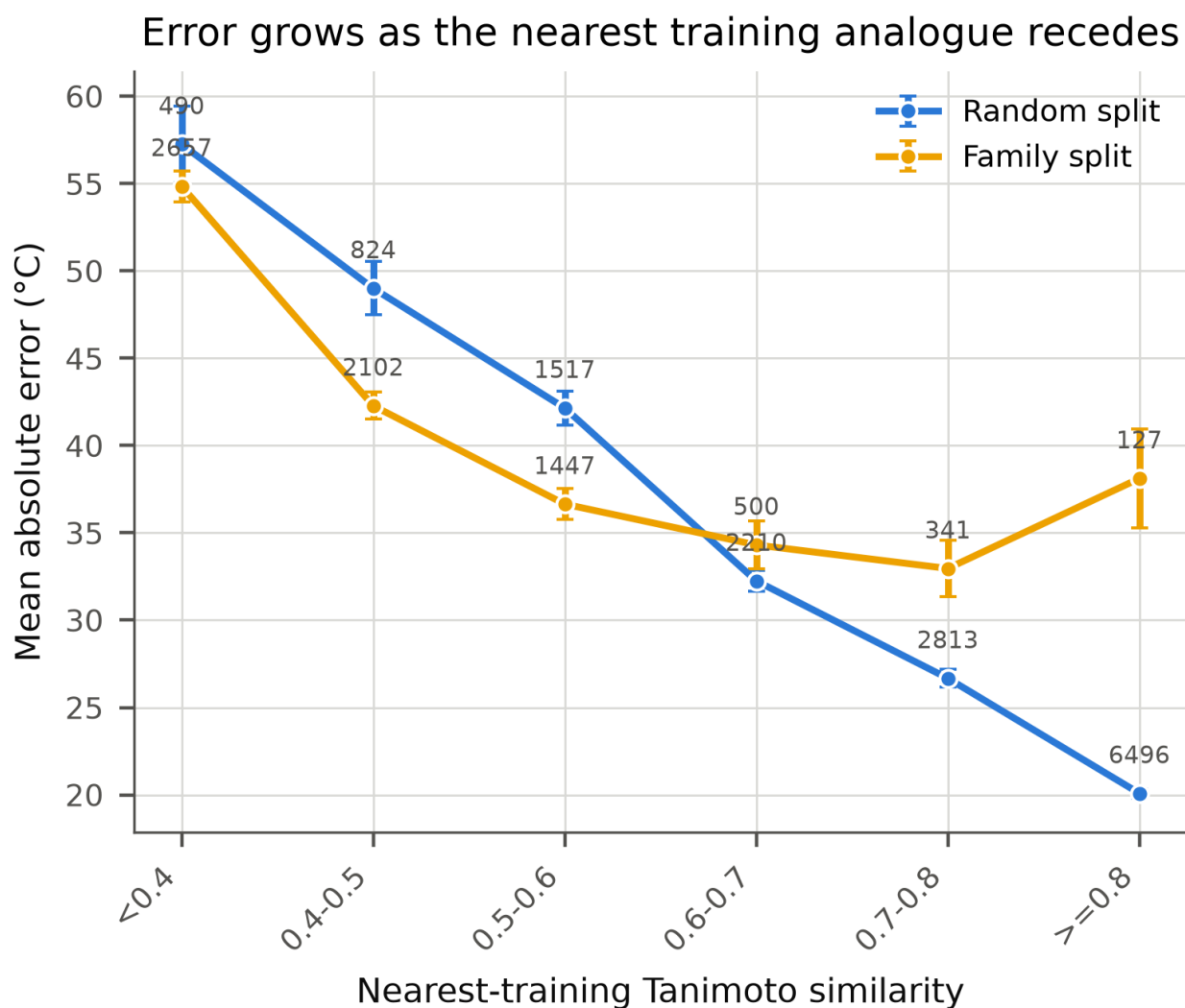
Both levers work, and to a similar degree (Table 9). Against a nominal 0.90 under family holdout, with the point model identical in all four arms:

| Calibration set | Quantile            | Coverage | Worst band   | Width |
|-----------------|---------------------|----------|--------------|-------|
| Random          | Global (SCP)        | 0.711    | 0.593        | 124 K |
| Random          | Novelty-conditioned | 0.883    | 0.763        | 208 K |
| Family-out      | Global (SCP)        | 0.855    | 0.762        | 187 K |
| Family-out      | Novelty-conditioned | 0.863    | <b>0.790</b> | 188 K |

Rebuilding the calibration set lifts plain split conformal from 0.711 to 0.855 without touching the quantile rule, the model or the representation — a change of calibration construction alone. It does so by widening intervals from 124 K to 187 K, which is the appropriate response: the residuals it now calibrates against are genuinely larger.

The two repairs are, however, largely **substitutes rather than complements**. Applying both gives 0.863 marginal coverage, no better than the novelty-conditioned quantile alone at 0.883, because they address the same root cause from opposite ends — that calibration residuals understate extrapolation error. The combination is nonetheless the best of the four on *conditional* coverage (0.790 in the worst similarity band against 0.763 and 0.762), so a practitioner who can afford the extra fits gains a little where it matters most.

The practical guidance follows the cost. Novelty-conditioned calibration is free: it reuses the same fitted model and calibration set, requiring only that similarity be computed. Family-out calibration requires refitting the model once per held-out training family — here eight additional fits per split. For most purposes the novelty-conditioned quantile alone is the better trade; where the worst-case band is what governs a decision, and compute is available, the two together are better still.



**Figure 7.** Mean absolute error against nearest-training similarity, for random and family-holdout splits. Group sizes are annotated.

# Appendix A. Result tables

Point accuracy by split regime and model

| regime   | model         | n splits | mae mean | mae sd | rmse mean | r2 mean  |
|----------|---------------|----------|----------|--------|-----------|----------|
| random   | hist_gbr      | 10       | 28.131   | 1.015  | 41.037    | 0.865    |
| random   | extra_trees   | 10       | 28.513   | 0.931  | 41.656    | 0.861    |
| random   | random_forest | 10       | 30.302   | 1.000  | 43.329    | 0.850    |
| random   | svr           | 10       | 30.875   | 0.896  | 45.482    | 0.835    |
| random   | median        | 10       | 93.786   | 1.650  | 112.146   | -0.00338 |
| scaffold | hist_gbr      | 10       | 33.807   | 7.764  | 45.950    | 0.792    |
| scaffold | svr           | 10       | 35.336   | 6.152  | 49.328    | 0.761    |
| scaffold | extra_trees   | 10       | 35.498   | 8.483  | 47.937    | 0.774    |
| scaffold | random_forest | 10       | 36.950   | 8.367  | 49.580    | 0.758    |
| scaffold | median        | 10       | 95.709   | 13.666 | 114.607   | -0.291   |
| cluster  | hist_gbr      | 10       | 44.466   | 10.373 | 58.590    | 0.538    |
| cluster  | extra_trees   | 10       | 48.705   | 11.271 | 62.946    | 0.462    |
| cluster  | random_forest | 10       | 48.791   | 11.807 | 63.327    | 0.458    |
| cluster  | svr           | 10       | 53.943   | 17.063 | 69.875    | 0.329    |
| cluster  | median        | 10       | 121.607  | 24.225 | 138.638   | -1.737   |
| family   | hist_gbr      | 20       | 46.816   | 15.589 | 61.130    | 0.322    |
| family   | extra_trees   | 20       | 48.866   | 16.214 | 62.673    | 0.286    |
| family   | random_forest | 20       | 49.338   | 15.781 | 63.305    | 0.269    |
| family   | svr           | 20       | 53.199   | 18.955 | 67.895    | 0.121    |
| family   | median        | 20       | 94.575   | 27.264 | 110.736   | -1.203   |

Conformal coverage, width and subgroup validity

| regime   | conformal           | coverage | worst similarity coverage | mean width | fallback |
|----------|---------------------|----------|---------------------------|------------|----------|
| random   | Mondrian-family     | 0.903    | 0.690                     | 135.567    | 0.062    |
| random   | Mondrian-similarity | 0.907    | 0.853                     | 129.075    | 0        |
| random   | Normalised SCP      | 0.902    | 0.741                     | 126.993    | 0        |
| random   | SCP                 | 0.904    | 0.650                     | 132.327    | 0        |
| scaffold | Mondrian-family     | 0.835    | 0.660                     | 129.014    | 0.096    |
| scaffold | Mondrian-similarity | 0.895    | 0.840                     | 154.832    | 0        |
| scaffold | Normalised SCP      | 0.825    | 0.683                     | 124.106    | 0        |
| scaffold | SCP                 | 0.834    | 0.624                     | 126.459    | 0        |
| cluster  | Mondrian-family     | 0.757    | 0.691                     | 141.548    | 0.196    |
| cluster  | Mondrian-similarity | 0.886    | 0.773                     | 232.226    | 0.128    |
| cluster  | Normalised SCP      | 0.711    | 0.660                     | 127.672    | 0        |
| cluster  | SCP                 | 0.730    | 0.674                     | 126.358    | 0        |
| family   | Mondrian-family     | 0.711    | 0.593                     | 123.664    | 1.000    |
| family   | Mondrian-similarity | 0.883    | 0.763                     | 207.635    | 0        |
| family   | Normalised SCP      | 0.712    | 0.598                     | 130.184    | 0        |
| family   | SCP                 | 0.711    | 0.593                     | 123.664    | 0        |

Matched-pair effects, with bootstrap confidence intervals

| model       | effect        | n families | mean   | ci low | ci high | wilcoxon p | n families positive |
|-------------|---------------|------------|--------|--------|---------|------------|---------------------|
| extra_trees | family_effect | 20         | 14.610 | 10.529 | 19.415  | 1.91e-06   | 20                  |
| extra_trees | size_effect   | 20         | 0.318  | -0.629 | 1.234   | 0.475      | 12                  |
| extra_trees | total_effect  | 20         | 14.928 | 11.159 | 19.605  | 1.91e-06   | 20                  |
| hist_gbr    | family_effect | 20         | 13.279 | 8.811  | 18.564  | 9.54e-06   | 18                  |
| hist_gbr    | size_effect   | 20         | 0.210  | -0.673 | 1.184   | 0.927      | 9                   |
| hist_gbr    | total_effect  | 20         | 13.489 | 9.350  | 18.390  | 1.91e-06   | 20                  |

**Family-holdout error and deterioration (histogram gradient boosting)**

| <b>group</b>     | <b>n</b> | <b>mae</b> | <b>r2</b> | <b>deterioration</b> | <b>mean nn<br/>similarity</b> |
|------------------|----------|------------|-----------|----------------------|-------------------------------|
| Polyphosphazenes | 118      | 84.839     | -1.339    | 3.016                | 0.304                         |
| Polydienes       | 89       | 72.493     | 0.135     | 2.577                | 0.368                         |
| Polysiloxanes    | 209      | 69.701     | -0.368    | 2.478                | 0.388                         |
| Polyureas        | 43       | 62.810     | 0.303     | 2.233                | 0.549                         |
| Polyolefins      | 172      | 56.150     | 0.476     | 1.996                | 0.403                         |
| Polyhalo-olefins | 56       | 51.667     | 0.141     | 1.837                | 0.333                         |
| Polyamides       | 982      | 48.193     | 0.516     | 1.713                | 0.489                         |
| Polyurethanes    | 189      | 47.890     | -0.192    | 1.702                | 0.477                         |
| Polyphenylenes   | 429      | 47.599     | 0.703     | 1.692                | 0.460                         |
| Polystyrenes     | 250      | 42.206     | 0.312     | 1.500                | 0.399                         |
| Polysulfides     | 60       | 42.020     | 0.746     | 1.494                | 0.516                         |
| Polyanhydrides   | 43       | 41.159     | 0.467     | 1.463                | 0.473                         |
| Polyimides       | 1707     | 37.759     | 0.584     | 1.342                | 0.406                         |
| Polysulfones     | 232      | 36.654     | 0.565     | 1.303                | 0.561                         |
| Polyimines       | 52       | 36.418     | 0.225     | 1.295                | 0.532                         |
| Polyacrylics     | 661      | 36.249     | 0.382     | 1.289                | 0.412                         |
| Polyesters       | 836      | 34.039     | 0.741     | 1.210                | 0.511                         |
| Polyvinyls       | 212      | 33.082     | 0.417     | 1.176                | 0.424                         |
| Polyethers       | 736      | 32.877     | 0.823     | 1.169                | 0.551                         |
| Polycarbonates   | 98       | 22.509     | 0.812     | 0.800                | 0.576                         |

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